Non-linear simultaneous equations

1

a $x=0, y=0$ and $x=4, y=16$

b $x=0, y=0$ and $x=-4, y=-8$

c $x=2, y=-2$ and $x=-3, y=-7$

d $x=0, y=-16$ and $x=1, y=-15$

e $x=3, y=1$ and $x=5, y=-15$

f $x=0, y=5$ and $x=-2, y=1$

g $x=1, y=3$ and $x=-2, y=6$

h $x=4, y=4$ and $x=-1, y=9$

i $x=-1, y=4$ and $x=11, y=16$

j $x=3, y=4$

k $x=-5, y=-28$ and $x=\dfrac{3}{2}, y=-2$

l $x=-2, y=2$ and $x=1, y=8$

2

a Example answer: $x \approx 2.3, y \approx 1.3$ and $x \approx -1.3, y \approx -2.3$

b Example answer: $x \approx 1.5, y \approx 5.5$ and $x \approx -2.5, y \approx 1.4$

Applications

3

a $x+y=60, y=x^2+4$

b $(7,53),(-8,68)$

4

a $x=-1, x=3$

b $x=3$ since the supply needs to be larger than zero.

c $y=18$

d The equilibrium quantity occurs when 3 thousand units are being produced, which has a corresponding price of 18 dollars per unit.